Mathematical Analysis - Week 2

Lectures 5-7 and Tutorial 1

Contents

Tutorial 1: proof and quantifiers 1
Lecture 5: continuous functions and intermediate values 3
Lecture 6: infinity and asymptotes 4
Lecture 7: sequences and completeness 13

Week 1 supplied the definition of a limit. Here we use it in three settings: continuity on an interval, behavior at infinity, and sequences. The same estimates keep reappearing; what changes is whether we choose a radius around a point or a threshold beyond which the estimate holds.

Tutorial 1: proof and quantifiers

1 Proof, logic, and limit language

1.1 Basic terminology

- An axiom is accepted as part of the framework in which we work.
- A theorem is established by proof; propositions and lemmas have the same logical status, with names that describe their role.
- A proof is a finite, logically valid argument using definitions, axioms and results already proved.
- In $P \Rightarrow Q$, P is the hypothesis and Q is the conclusion. The converse $Q \Rightarrow P$ requires its own proof.

1.2 The ε-δ definition

Here δ controls how close the input x is to x_0, while ε prescribes how close $f(x)$ must be to ℓ.

A standard proof has the following shape:

1. Let $\varepsilon > 0$ be arbitrary.
2. Choose an explicit $\delta > 0$, usually after estimating $|f(x) - \ell|$ in terms of $|x - x_0|$.
3. Assume $0 < |x - x_0| < \delta$ and verify $|f(x) - \ell| < \varepsilon$.

Working backward helps find a suitable radius, but the written proof should then run forward from that choice. A list of inequalities ending in a proposed δ is not yet the verification.

1.3 Direct proof, contradiction, and contraposition

Direct proof To prove a statement of the form $P \Rightarrow Q$, assume that P is true and use definitions, axioms, and previously proved results until Q follows. This is the default method for most ε-δ arguments.

Proof by contradiction To prove a statement S:

1. Assume its negation, $\neg S$.
2. Derive a contradiction, such as $R \wedge \neg R$, an impossible inequality, or a conflict with a known theorem.

3. Conclude that $\neg S$ is false, so S is true.

For a uniqueness statement, assume that two distinct objects both satisfy the defining property. For a nonexistence statement, assume that the object exists and use its required properties to obtain a contradiction.

Example: irrationality of $\sqrt{2}$ Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$, where p, q are positive integers with no common factor. Squaring gives $p^2 = 2q^2$.

Thus p^2 is even, so p is even. Write $p = 2r$. Substitution gives

$$4r^2 = 2q^2 \quad \Longrightarrow \quad q^2 = 2r^2.$$

Therefore q is also even. Hence p and q have the common factor 2, contradicting the assumption that p/q is in lowest terms. Consequently, $\sqrt{2} \notin \mathbb{Q}$.

Proof by contraposition To prove $P \Rightarrow Q$, it is enough to prove the logically equivalent statement $\neg Q \Rightarrow \neg P$.

This method is especially useful when the negation of the conclusion gives more concrete information than the conclusion itself.

1.4 Mathematical induction

Let $P(n)$ be a statement for $n \in \mathbb{Z}_{>0}$. The principle of mathematical induction says that if:

1. **Base case:** $P(1)$ is true;
2. **Inductive step:** for every positive integer n, assuming $P(n)$ is true implies that $P(n+1)$ is true;

then $P(n)$ is true for every positive integer n.

In the inductive step, clearly state the **induction hypothesis** $P(n)$ before proving $P(n+1)$. Do not assume the statement that is being proved.

Example: sum of the first n positive integers We prove that

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \qquad (n \geq 1).$$

For $n = 1$, both sides equal 1. Now assume the formula holds for $n = k$: $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$.

Then

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus, the statement for k implies the statement for $k+1$, so the formula holds for every $n \geq 1$.

Strong induction The principle of strong induction replaces the induction hypothesis $P(n)$ by $P(1), P(2), \ldots, P(n) \quad \Longrightarrow \quad P(n+1)$.

It is useful when the case $n+1$ depends on several earlier cases rather than only on the immediately preceding case.

Ordinary induction, strong induction, and the well-ordering principle are equivalent principles. The well-ordering principle states that every nonempty subset of $\mathbb{Z}_{>0}$ has a least element.

Lecture 5: continuous functions and intermediate values

Composition and change of variable

Continuity of a composition. If $g(x) \to b$ as $x \to c$, and f is continuous at b, with $f(g(x))$ defined near c, then

$$\lim_{x \to c} f(g(x)) = f(b).$$

Given $\varepsilon > 0$, continuity of f gives an input tolerance $\eta > 0$. The limit of g then supplies δ such that $0 < |x - c| < \delta$ implies $|g(x) - b| < \eta$, which in turn implies $|f(g(x)) - f(b)| < \varepsilon$. If $g(x) = b$, continuity still covers that value.

If we assume only $f(t) \to A$ as $t \to b$, rather than continuity at b, we need $g(x) \neq b$ on a punctured neighborhood. For example, take $f(t) = 1$ for $t \neq 0$, $f(0) = 0$, and $g(x) = 0$. Then $f(t) \to 1$ at zero, but $f(g(x)) = 0$.

A two-way substitution requires more than $u(t) \to c$. If u is a continuous bijection between neighborhoods of t_0 and c, with $u(t_0) = c$ and continuous inverse, the two compositions give

$$\lim_{x \to c} f(x) = A \quad \Longleftrightarrow \quad \lim_{t \to t_0} f(u(t)) = A.$$

Squaring near zero is not such a change of variable on a two-sided interval: it loses the negative side.

The Intermediate Value Theorem

If f is continuous on $[a, b]$, every value between $f(a)$ and $f(b)$ is attained at some point in $[a, b]$. In particular, opposite signs at the endpoints imply a zero in (a, b).

The theorem gives existence, not uniqueness. A sign change is sufficient, not necessary: x^2 has a zero without changing sign. Continuity is essential; a step function can jump across zero without taking the value zero.

Exercise 1.1.26

Any polynomial of odd degree with real coefficients has at least one real root.

Proof

Let

$$P(x) = \sum_{k=0}^{n} a_k x^k$$

be a polynomial of degree n with real coefficients, where n is an odd integer and $a_n \neq 0$. Without loss of generality, assume that $a_n > 0$. If $a_n < 0$, we can apply the same argument to $-P$.

Let $C = \max_{0 \leq k \leq n-1} |a_k|$.

For $|x| \geq 1$, we have

$$\begin{aligned} |P(x) - a_n x^n| &= \left| \sum_{k=0}^{n-1} a_k x^k \right| \\ &\leq C \sum_{k=0}^{n-1} |x|^k \\ &\leq Cn|x|^{n-1} = \frac{Cn}{|a_n||x|} |a_n x^n|. \end{aligned}$$

Choose $R > \max\left\{1, \frac{2Cn}{|a_n|}\right\}$.

Then, whenever $|x| \geq R$, $|P(x) - a_n x^n| < \frac{1}{2}|a_n x^n|$.

Since n is odd, $R^n > 0$ and $(-R)^n < 0$. For $x = R$, we obtain

$$\begin{aligned} P(R) &\geq a_n R^n - |P(R) - a_n R^n| \\ &> a_n R^n - \frac{1}{2}|a_n R^n| = \frac{1}{2} a_n R^n > 0. \end{aligned}$$

For $x = -R$, we obtain

$$\begin{aligned} P(-R) &\leq a_n (-R)^n + |P(-R) - a_n (-R)^n| \\ &< a_n (-R)^n + \frac{1}{2}|a_n (-R)^n| = \frac{1}{2} a_n (-R)^n < 0. \end{aligned}$$

Thus, $P(-R) < 0 < P(R)$.

Because P is continuous on $[-R, R]$, the Intermediate Value Theorem gives some $c \in [-R, R]$ such that $P(c) = 0$.

Therefore, P has at least one real root. $\square$

Proof of the Intermediate Value Theorem

Let f be continuous on $[a, b]$, and suppose first that $f(a) < y_0 < f(b)$.

Define $S = \{x \in [a, b] : f(x) \leq y_0\}$.

The set S is nonempty because $a \in S$, and it is bounded above by b. By the completeness of $\mathbb{R}$, the supremum $c = \sup S$ exists. Continuity at a gives a short interval $[a, a + \eta]$ on which $f < y_0$; continuity at b gives a short interval $[b - \eta, b]$ on which $f > y_0$, after reducing $\eta > 0$ if needed. Thus $a + \eta \leq c \leq b - \eta$, so $a < c < b$. This is the endpoint step needed before using a two-sided neighborhood of c.

We claim that $f(c) = y_0$. Suppose first that $f(c) < y_0$. By continuity of f at c, there exists $\delta > 0$ such that $|x - c| < \delta \implies f(x) < y_0$.

Choosing $x > c$ sufficiently close to c gives $x \in S$, contradicting the fact that c is an upper bound of S.

On the other hand, suppose that $f(c) > y_0$. By continuity of f at c, there exists $\delta > 0$ such that $|x - c| < \delta \implies f(x) > y_0$.

Since $c = \sup S$, there is some $x \in S$ with $c - \delta < x \leq c$. This contradicts $f(x) > y_0$.

Therefore neither $f(c) < y_0$ nor $f(c) > y_0$ is possible, and hence $f(c) = y_0$.

If $y_0 = f(a)$ or $y_0 = f(b)$, the conclusion is immediate by taking $c = a$ or $c = b$. If $f(b) < f(a)$, apply the same argument to $-f$. Therefore, for every value y_0 between $f(a)$ and $f(b)$, there exists $c \in [a, b]$ such that $f(c) = y_0$.

Lecture 6: infinity and asymptotes

Limits Involving Infinity

Finite limits as $x \to +\infty$

We say that

$$\lim_{x \to +\infty} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $M > 0$ such that $x > M \quad \Longrightarrow \quad |f(x) - L| < \varepsilon$.
In words, $f(x)$ becomes arbitrarily close to L when x is sufficiently large.
Similarly,

$$\lim_{x \to -\infty} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $M < 0$ such that $x < M \quad \Longrightarrow \quad |f(x) - L| < \varepsilon$.

Examples

For the reciprocal function,

$$\lim_{x \to +\infty} \frac{1}{x} = 0, \qquad \lim_{x \to -\infty} \frac{1}{x} = 0.$$

Indeed, if $|x| > 1/\varepsilon$, then $\left|\frac{1}{x} - 0\right| = \frac{1}{|x|} < \varepsilon$.
For a rational function whose numerator has lower degree than its denominator,

$$\lim_{x \to \pm\infty} \frac{P(x)}{Q(x)} = 0, \qquad \deg P < \deg Q.$$

Infinite limits as $x \to c$

We write

$$\lim_{x \to c} f(x) = +\infty$$

if, for every $A > 0$, there exists $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \Longrightarrow \quad f(x) > A.$$

Similarly,

$$\lim_{x \to c} f(x) = -\infty$$

if, for every $B > 0$, there exists $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \Longrightarrow \quad f(x) < -B.$$

For example,

$$\lim_{x \to 0} \frac{1}{x^2} = +\infty,$$

because $1/x^2$ can be made larger than any prescribed positive number by choosing x sufficiently close to 0, with $x \neq 0$.

Infinite limits as $x \to \pm\infty$

The notation

$$\lim_{x\to+\infty} f(x) = +\infty$$

means that for every $A > 0$, there exists $M > 0$ such that $x > M \quad \Longrightarrow \quad f(x) > A$.
The other cases are defined analogously. For instance, if n is odd and $a_n > 0$, then

$$\lim_{x\to+\infty} a_n x^n = +\infty, \qquad \lim_{x\to-\infty} a_n x^n = -\infty.$$

Exercise 1.1.30

(a)

Prove, using the N-δ language, that

$$\lim_{x\to0^+} \frac{1}{\sqrt[3]{x}} = +\infty.$$

(b)

Find the limit

$$\lim_{x\to+\infty} \left(\sqrt{x+1} - \sqrt{x-1}\right),$$

and prove your result using the ε-N language.

Solution

(a) Let $N > 0$ be arbitrary. Choose $\delta = \frac{1}{N^3}$.
If $0 < x < \delta$, then $\sqrt[3]{x} < \sqrt[3]{\delta} = \frac{1}{N}$.
Therefore, $\frac{1}{\sqrt[3]{x}} > \frac{1}{\sqrt[3]{\delta}} = N$.
Hence, for every $N > 0$, there exists $\delta > 0$ such that

$$0 < x < \delta \quad \Longrightarrow \quad \frac{1}{\sqrt[3]{x}} > N.$$

Thus,

$$\boxed{\lim_{x\to0^+} \frac{1}{\sqrt[3]{x}} = +\infty}.$$

(b) Let $f(x) = \sqrt{x+1} - \sqrt{x-1}$.
For $x \geq 1$, rationalizing gives

$$f(x) = \frac{(x+1)-(x-1)}{\sqrt{x+1}+\sqrt{x-1}} = \frac{2}{\sqrt{x+1}+\sqrt{x-1}}.$$

Since $\sqrt{x+1} + \sqrt{x-1} \to +\infty$ as $x \to +\infty$, we obtain

$$\lim_{x\to+\infty} f(x) = 0.$$

To prove this using the ε-N language, let $\varepsilon > 0$ be given. Choose $N = 1 + \frac{1}{\varepsilon^2}$.
If $x > N$, then $x > 1$ and

$$0 < f(x) = \frac{2}{\sqrt{x+1} + \sqrt{x-1}} \leq \frac{1}{\sqrt{x-1}}.$$

Moreover, $x > N$ implies $x - 1 > 1/\varepsilon^2$, so

$$|f(x) - 0| = f(x) \leq \frac{1}{\sqrt{x-1}} < \varepsilon.$$

Therefore, for every $\varepsilon > 0$, there exists $N > 0$ such that

$$x > N \quad \Longrightarrow \quad \left|\sqrt{x+1} - \sqrt{x-1}\right| < \varepsilon.$$

Hence,

$$\boxed{\lim_{x \to +\infty} \left(\sqrt{x+1} - \sqrt{x-1}\right) = 0}.$$

Exercise 1.1.28

All the Limit Laws remain true when we replace

$$\lim_{x \to c}$$

by

$$\lim_{x \to +\infty} \quad \text{or} \quad \lim_{x \to -\infty}.$$

For example, if

$$\lim_{x \to +\infty} f(x) = L, \qquad \lim_{x \to +\infty} g(x) = M,$$

then

$$\lim_{x \to +\infty} (f(x) + g(x)) = L + M,$$

and, provided $M \neq 0$,

$$\lim_{x \to +\infty} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

The same statements hold as $x \to -\infty$.

Asymptotes

The graph of $y = f(x)$ on an interval I is $\{(x, y) : x \in I,\ y = f(x)\}$.
A line is an asymptote when the graph approaches it along an unbounded branch. For horizontal and oblique lines we let x tend to an infinity. For a vertical line $x = a$, we require a one-sided infinite limit of f at a. Merely having $x \to a$ is not enough: an ordinary point of a curve is not a vertical asymptote.

Horizontal asymptotes

A line $y = b$ is a horizontal asymptote of $y = f(x)$ as $x \to +\infty$ if

$$\lim_{x \to +\infty} f(x) = b.$$

Similarly, $y = b$ is a horizontal asymptote as $x \to -\infty$ if

$$\lim_{x \to -\infty} f(x) = b.$$

Example 1.1.32 Find the horizontal asymptotes of $f(x) = \frac{\sin x + \cos x}{1+x^2}$, and

$$g(x) = \sqrt{1 - \frac{1}{x^2}}, \qquad |x| \geq 1.$$

Since $|\sin x + \cos x| \leq 2$,

$$\left| \frac{\sin x + \cos x}{1 + x^2} \right| \leq \frac{2}{1 + x^2} \longrightarrow 0.$$

Thus, $y = 0$ is a horizontal asymptote of f as $x \to \pm\infty$.
For g,

$$\lim_{x \to \pm\infty} g(x) = \sqrt{1 - \lim_{x \to \pm\infty} \frac{1}{x^2}} = 1.$$

Thus, $y = 1$ is a horizontal asymptote of g as $x \to \pm\infty$.

Sketches of horizontal asymptotes

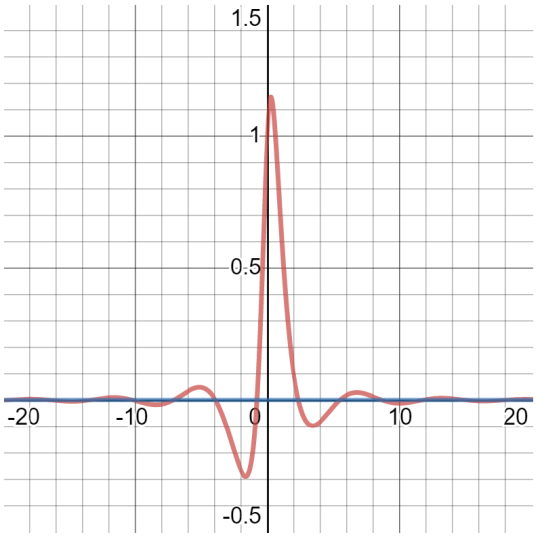


Figure: $f = \frac{\sin x + \cos x}{1+x^2}$

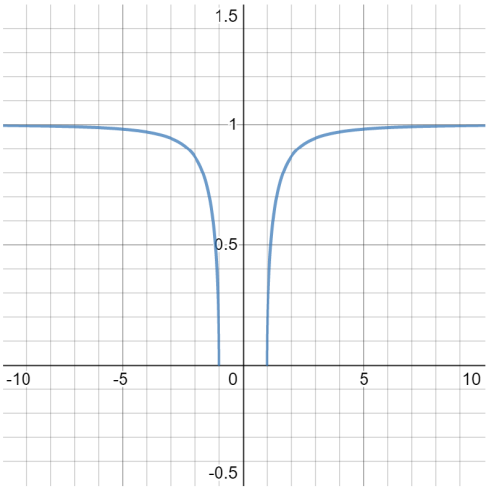


Figure: $g = \sqrt{1 - \frac{1}{x^2}}$

Vertical asymptotes

A line $x = a$ is a vertical asymptote of $y = f(x)$ as $x \to a^+$ if

$$\lim_{x \to a^+} f(x) = +\infty \quad \text{or} \quad \lim_{x \to a^+} f(x) = -\infty.$$

The analogous definition applies as $x \to a^-$. In the notation $\pm\infty$, the sign may be either $+$ or $-$.

Example 1.1.33 Consider $h(x) = \sqrt{1 - \frac{\sin x}{x^2}}$, on its domain $(-\infty, 0) \cup [a, +\infty)$, where the boundary point $a > 0$ is determined by $a^2 = \sin a$, where $0 < a < 1$. As $x \to 0^-$, we have $\sin x \sim x$, and therefore

$$1 - \frac{\sin x}{x^2} \sim 1 - \frac{1}{x} \longrightarrow +\infty.$$

Consequently,

$$\lim_{x \to 0^-} h(x) = +\infty,$$

so $x = 0$ is a vertical asymptote. Near $x = a$ from the right, the expression under the square root approaches 0, so $x = a$ is not a vertical asymptote.

This example emphasizes that a domain endpoint is not automatically a vertical asymptote: the function itself must tend to $\pm\infty$.

Sketches of vertical asymptotes

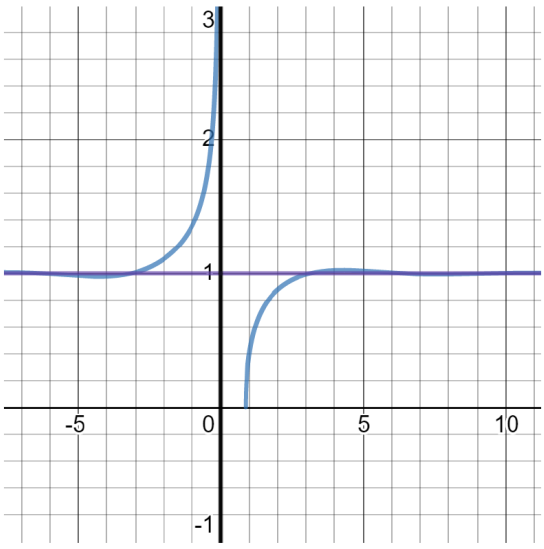


Figure: $f = \sqrt{1 - \frac{\sin x}{x^2}}$

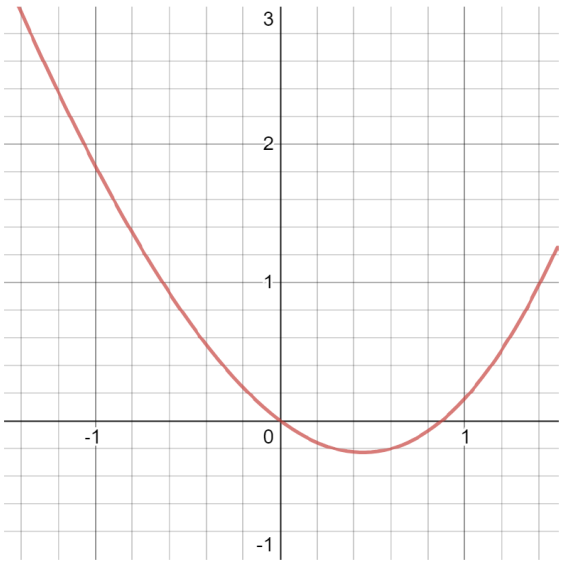


Figure: $y = x^2 - \sin x$

Oblique asymptotes

A line

$$y = ax + b, \qquad a \neq 0,$$

is an oblique asymptote of $y = f(x)$ as $x \to +\infty$ if

$$\lim_{x\to+\infty}[f(x)-ax-b]=0.$$

The definition as $x\to-\infty$ is analogous.

When the limits exist, the coefficients can be found from

$$a=\lim_{x\to\pm\infty}\frac{f(x)}{x},\qquad b=\lim_{x\to\pm\infty}[f(x)-ax].$$

Example 1.1.34 Find the oblique asymptote of $f(x)=\frac{x^2-3}{2x-4}$.

Polynomial division gives

$$\begin{aligned}f(x)&=\frac{x^2-4+1}{2(x-2)}\\&=\frac{1}{2}(x+2)+\frac{1}{2(x-2)}.\end{aligned}$$

Therefore,

$$f(x)-\left(\frac{x}{2}+1\right)=\frac{1}{2(x-2)}\longrightarrow 0\qquad(x\to\pm\infty).$$

The oblique asymptote is $y=\frac{x}{2}+1$.

Also, $x=2$ is a vertical asymptote.

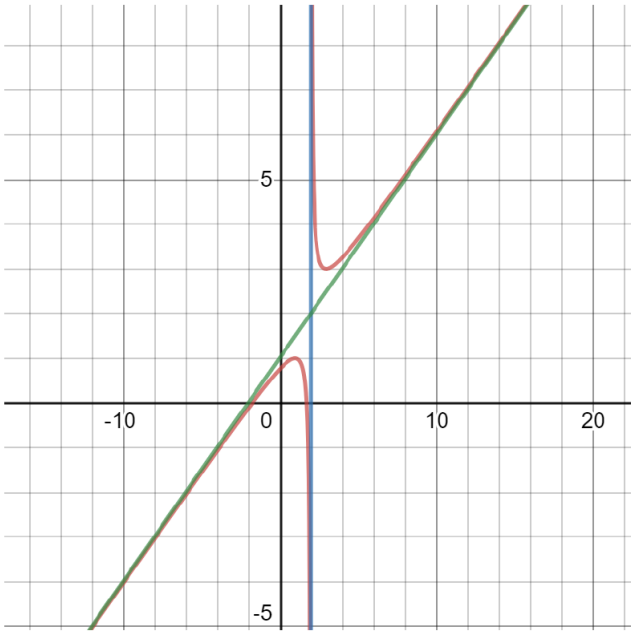


Figure: $f=\frac{x^2-3}{2x-4}$

Example 1.1.35 Let $f(x)=\frac{x^3}{(x+2)(x-1)}$.

The vertical asymptotes are

$$x=-2,\qquad x=1.$$

For the oblique asymptote,

$$\frac{f(x)}{x} = \frac{x^2}{(x+2)(x-1)} \longrightarrow 1,$$

and

$$f(x) - x = \frac{-x^2 + 2x}{(x+2)(x-1)} \longrightarrow -1.$$

Hence the oblique asymptote is $y = x - 1$ as $x \to \pm\infty$.

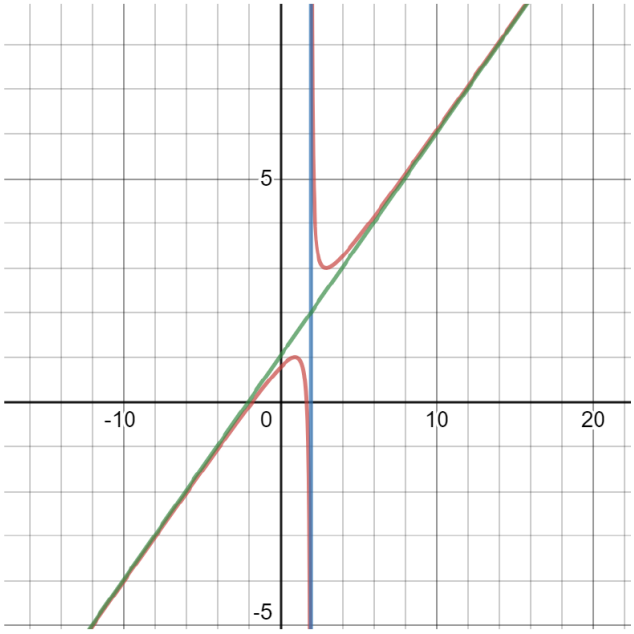


Figure: $f = \frac{x^2-3}{2x-4}$

Exercise 1.1.36

Let $f(x) = \frac{1}{x}\sin\frac{1}{x}$.

Does the graph of f have a vertical asymptote at $x = 0$? Explain your answer using the definition of a vertical asymptote.

Since $\left|\frac{1}{x}\sin\frac{1}{x}\right| \leq \frac{1}{|x|}$, this estimate alone does not determine an infinite limit. To see the oscillation explicitly, define

$$x_n = \frac{1}{\frac{\pi}{2} + 2\pi n}, \qquad y_n = \frac{1}{\frac{3\pi}{2} + 2\pi n}.$$

Then $x_n, y_n \to 0^+$ and $f(x_n) = \frac{\pi}{2} + 2\pi n \longrightarrow +\infty$, while $f(y_n) = -\left(\frac{3\pi}{2} + 2\pi n\right) \longrightarrow -\infty$.

Thus, as $x \to 0^+$, the function tends neither to $+\infty$ nor to $-\infty$. Applying the same argument to $-x_n$ and $-y_n$ gives the same conclusion as $x \to 0^-$. Therefore, $x = 0$ is not a vertical asymptote in the usual one-sided sense.

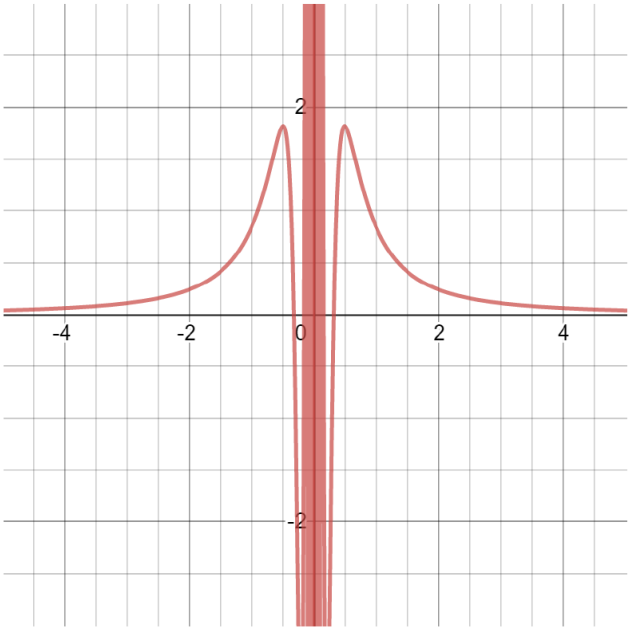


Figure: $f = \frac{1}{x}\sin\frac{1}{x}$

More Exercises

Find the limits and use the ε-N language to prove your results.

(a)

$$\lim_{x\to+\infty} \sqrt{x}\left(\sqrt{x+1}-\sqrt{x-1}\right).$$

(b)

$$\lim_{x\to+\infty} \frac{\sqrt{x+\sqrt{x+\sqrt{x}}}}{\sqrt{x+1}}.$$

Solution to (a). Rationalizing and dividing the denominator by $\sqrt{x}$ gives $A(x) = \frac{2}{\sqrt{1+1/x}+\sqrt{1-1/x}} \longrightarrow 1$.

For $x \geq 2$, write $S = \sqrt{1+1/x}+\sqrt{1-1/x} \geq 1$. Rationalizing each difference from 1 shows $|S-2| \leq 2/x$, so $|A(x)-1| \leq 2/x$. The choice $N = \max\{2, 2/\varepsilon\}$ verifies the limit directly.

Solution to (b). The denominator is outside the outer radical, as in Lecture 6, p.32. Put

$$B(x) = \sqrt{\frac{x+\sqrt{x+\sqrt{x}}}{x+1}}, \qquad x \geq 1.$$

Then $B(x) \to 1$. To obtain a threshold rather than only a formal calculation, use

$$|B(x)-1| = \frac{\sqrt{x+\sqrt{x}}-1}{(x+1)(B(x)+1)} \leq \frac{\sqrt{2x}}{x+1} \leq \frac{\sqrt{2}}{\sqrt{x}}.$$

Thus $N = \max\{1, 2/\varepsilon^2\}$ works. Moving the denominator inside the radical would change the problem.

Lecture 7: sequences and completeness

1.7 Sequences and limits

Definition 1.1.12 (Limit of a sequence)

A sequence $\{a_n\}$ converges to L if for every $\varepsilon > 0$ there exists an integer N such that $n > N \implies |a_n - L| < \varepsilon$.

If no such number L exists, we say that the sequence diverges.

We write

$$\lim_{n\to\infty} a_n = L, \qquad \text{or} \qquad a_n \to L.$$

Exercise 1.1.39

Show that if a sequence converges, then the limit is unique.

Proof Suppose that

$$a_n \to L_1, \qquad a_n \to L_2.$$

Let $\varepsilon > 0$ be arbitrary. Then there exist N_1, N_2 such that $n > N_1 \Rightarrow |a_n - L_1| < \varepsilon/2$, and $n > N_2 \Rightarrow |a_n - L_2| < \varepsilon/2$.

For $n > \max\{N_1, N_2\}$,

$$|L_1 - L_2| \le |a_n - L_1| + |a_n - L_2| < \varepsilon.$$

Since this holds for every $\varepsilon > 0$, we must have $L_1 = L_2$.

Hence the limit is unique.

Example 1.1.40

Investigate the convergence or divergence of the following sequences.

(a) $(-1)^n$ This sequence alternates between -1 and 1, so it does not approach any single number. Therefore, $(-1)^n$ diverges.

(b) $\dfrac{\sin n - n^2}{n^3 + 1}$ Since

$$\frac{\sin n - n^2}{n^3 + 1} = \frac{\sin n}{n^3 + 1} - \frac{n^2}{n^3 + 1},$$

and both terms go to 0, we have

$$\lim_{n\to\infty} \frac{\sin n - n^2}{n^3 + 1} = 0.$$

(c) $\dfrac{(-1)^{[\sqrt{n}]}}{n}$ Here $[\sqrt{n}]$ is the integer part of $\sqrt{n}$, and the numerator is always ± 1. Since dividing by n makes the magnitude go to 0,

$$\lim_{n\to\infty} \frac{(-1)^{[\sqrt{n}]}}{n} = 0.$$

Bounded sequences

Definition

A sequence $\{a_n\}$ is called:

- bounded above if there exists a constant M such that $a_n \leq M$ for all n;
- bounded below if there exists a constant m such that $a_n \geq m$ for all n;
- bounded if there exists a constant $C > 0$ such that $|a_n| \leq C$ for all n.

Equivalently, $\{a_n\}$ is bounded iff there exist m, M with

$$m \leq a_n \leq M \qquad \forall n \geq 1.$$

Theorem

Every convergent sequence is bounded.

Proof If $a_n \to L$, then for $\varepsilon = 1$ there exists N such that $n > N \Rightarrow |a_n - L| < 1$.
Thus for all $n > N$, $|a_n| \leq |L| + 1$.
Let $C = \max\{|a_1|, \ldots, |a_N|, |L| + 1\}$.
Then $|a_n| \leq C$ for all n, so the sequence is bounded.

Divergence to infinity

Definition

We say that $a_n \to +\infty$ if for every real number M there exists an integer N such that $n \geq N \Rightarrow a_n > M$.

We say that $a_n \to -\infty$ if for every real number $m > 0$ there exists an integer N such that $n \geq N \Rightarrow a_n < -m$.

Example 1.1.41

Investigate the behavior of the following sequences as $n \to \infty$.

(a) $(-1)^n n$ This sequence alternates sign and grows in magnitude, so it is unbounded and does not converge to a finite limit. It also does not tend to $+\infty$ or $-\infty$ in the usual sense because the signs alternate.

(b) $a^n/n!$ **with** $a > 0$ For fixed $a > 0$, set $b_n = a^n/n!$. For n large enough, $\frac{b_{n+1}}{b_n} = \frac{a}{n+1} \leq \frac{1}{2}$.
Hence $b_{N+k} \leq b_N 2^{-k} \to 0$. This gives a proof of the statement that a factorial eventually dominates a fixed exponential: $\frac{a^n}{n!} \to 0$.

(c) $n^n/n!$ Since

$$\frac{n^n}{n!} = \prod_{k=1}^{n} \frac{n}{k},$$

the factors with $k \leq \lfloor n/2 \rfloor$ are at least 2, and all remaining factors are at least 1. Consequently $\frac{n^n}{n!} \geq 2^{\lfloor n/2 \rfloor} \longrightarrow +\infty$.
In particular, $\frac{n^n}{n!} \to +\infty$.

Limit laws for sequences

If

$$\lim_{n\to\infty} a_n = A, \qquad \lim_{n\to\infty} b_n = B,$$

then the following hold:

1. $\lim_{n\to\infty} (a_n + b_n) = A + B$;
2. $\lim_{n\to\infty} (a_n - b_n) = A - B$;
3. $\lim_{n\to\infty} (ka_n) = kA$ for any constant k;
4. $\lim_{n\to\infty} (a_n b_n) = AB$;
5. if $B \neq 0$, then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}.$$

Exercise 1.1.43

Let $\{a_n\}$ and $\{b_n\}$ be convergent sequences and assume $a_n \geq b_n$ for all sufficiently large n. Prove that

$$\lim_{n\to\infty} a_n \geq \lim_{n\to\infty} b_n.$$

Proof

Let

$$A = \lim_{n\to\infty} a_n, \qquad B = \lim_{n\to\infty} b_n.$$

Suppose, for contradiction, that $A < B$. Set $\varepsilon = \frac{B-A}{2} > 0$.

Then there exists N such that for all $n > N$,

$$|a_n - A| < \varepsilon, \qquad |b_n - B| < \varepsilon.$$

Hence for $n > N$, $a_n < A + \varepsilon = \frac{A+B}{2}$, and $b_n > B - \varepsilon = \frac{A+B}{2}$.

So $a_n < b_n$ for all sufficiently large n, contradicting the assumption that $a_n \geq b_n$ eventually. Therefore, $A \geq B$.

Exercise 1.1.43 (continued)

Assume $a_n \leq b_n \leq c_n$ for all $n \geq N_0$ and

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L.$$

Then

$$\lim_{n\to\infty} b_n = L.$$

Proof

Let $\varepsilon > 0$ be given. Since $a_n \to L$ and $c_n \to L$, there exist N_1, N_2 such that $n > N_1 \Rightarrow |a_n - L| < \varepsilon$, and $n > N_2 \Rightarrow |c_n - L| < \varepsilon$.

Let $N = \max\{N_0, N_1, N_2\}$. Then for all $n > N$, $L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon$.

Hence $|b_n - L| < \varepsilon$, for all $n > N$. Therefore, $b_n \to L$.

This is the sequence version of the Sandwich Theorem.

Exercise 1.1.44

Let $\{a_n\}$ be a sequence with $a_n \to L$, and let f be a function continuous at L and defined at all a_n. Then $f(a_n) \to f(L)$.

Proof

Let $\varepsilon > 0$ be given. Since f is continuous at L, there exists $\delta > 0$ such that $|x - L| < \delta \Rightarrow |f(x) - f(L)| < \varepsilon$.

Because $a_n \to L$, there exists N such that $n > N \Rightarrow |a_n - L| < \delta$.

Therefore, for all $n > N$, $|f(a_n) - f(L)| < \varepsilon$.

So $f(a_n) \to f(L)$.

Proposition 1.1.13 (function limits and sequence limits)

Let I be an interval containing x_0 in its interior and let f be defined on $I \setminus \{x_0\}$. Then

$$\lim_{x \to x_0} f(x) = A$$

exists if and only if for every sequence $\{a_n\} \subset I \setminus \{x_0\}$ with $a_n \to x_0$, the limit

$$\lim_{n \to \infty} f(a_n)$$

exists and equals A.

Proof

- If $f(x) \to A$, fix $\varepsilon > 0$ and take its limit radius δ. Since $a_n \to x_0$ with $a_n \neq x_0$, eventually $0 < |a_n - x_0| < \delta$. Hence $|f(a_n) - A| < \varepsilon$.
- Conversely, suppose the function limit were not A. Negating the definition supplies $\varepsilon_0 > 0$ such that every radius contains a point violating the error bound. For each n, choose x_n with $0 < |x_n - x_0| < 1/n$ and $|f(x_n) - A| \geq \varepsilon_0$. Then $x_n \to x_0$ but $f(x_n)$ does not converge to A, a contradiction.

The version in the lecture says only that $f(a_n)$ converges for every admissible sequence. The common value then follows too: if two sequences gave different limits, interleaving them would produce an admissible sequence whose image does not converge.

Thus, the limit of a function can be tested using sequences.

Exercise 1.1.45

Let f be a function defined on $[M, \infty)$. Prove:

(a)

$$\lim_{x\to\infty} f(x) = L$$

if and only if for every sequence $a_n \to \infty$,

$$\lim_{n\to\infty} f(a_n) = L.$$

(b)

$$\lim_{x\to\infty} f(x) = +\infty$$

if and only if for every sequence $a_n \to \infty$,

$$\lim_{n\to\infty} f(a_n) = +\infty.$$

Proof. The forward implications follow directly from the threshold definitions. If the finite limit fails, there are $\varepsilon_0 > 0$ and points $x_n > \max\{M, n\}$ with $|f(x_n) - L| \geq \varepsilon_0$. If the infinite limit fails, there are a fixed bound $A > 0$ and points $x_n > \max\{M, n\}$ with $f(x_n) \leq A$. In either case $x_n \to \infty$ contradicts the corresponding hypothesis about all sequences.

1.8 Least upper bound and greatest lower bound

Definition

Let $\{a_n\}$ be a sequence bounded above. A number L is called the least upper bound, or supremum, of $\{a_n\}$ if:

1. $a_n \leq L$ for all n;
2. if M is any upper bound, then $L \leq M$.

We write $L = \sup_{n\geq 1} a_n$.

Similarly, if $\{a_n\}$ is bounded below, a number l is called the greatest lower bound, or infimum, if:

1. $a_n \geq l$ for all n;
2. if m is any lower bound, then $l \geq m$.

We write $l = \inf_{n\geq 1} a_n$.

Remarks

- For the supremum: for every $\varepsilon > 0$, there exists n such that

$$a_n > L - \varepsilon.$$

- For the infimum: for every $\varepsilon > 0$, there exists n such that

$$a_n < l + \varepsilon.$$

Theorem 1.1.15

Any sequence bounded above has a supremum, and any sequence bounded below has an infimum.

This is the completeness property of the real numbers.

Example 1.1.47

Find the infimum and supremum of the following sequences.

(a) $\{(-1)^n/n\}_{n\geq 1}$

$$\inf = -1, \qquad \sup = \frac{1}{2}.$$

The first term is -1, so an infimum of $-1/2$ would already fail to be a lower bound. The odd-indexed terms increase from -1 toward zero; the even-indexed terms decrease from $1/2$ toward zero. Both extremal values are attained.

(b) $\{(-1)^n n\}_{n\geq 1}$

This sequence is unbounded above and below, so

$$\sup = +\infty, \qquad \inf = -\infty.$$

(c) $\{n^{(-1)^n}\}_{n\geq 1}$

This is a power, not the product in (b). Even indices give $a_{2k} = 2k$, whereas odd indices give $a_{2k-1} = 1/(2k-1)$. Therefore

$$\sup = +\infty, \qquad \inf = 0.$$

There is no minimum: every term is positive, but the odd subsequence tends to zero. The product $n(-1)^n$ in the original transcription is the same sequence as (b), whose extended bounds are $-\infty$ and $+\infty$.

(d) $a_1 = \sqrt{2}$, $a_{n+1} = \sqrt{2 + a_n}$

Induction gives $0 < a_n < 2$: if $a_n < 2$, then $a_{n+1} = \sqrt{2 + a_n} < 2$. Also $a_2 > a_1$, and monotonicity of the square root propagates $a_{n+1} > a_n$. An increasing, bounded sequence converges: if $L = \sup\{a_n\}$, then for every $\varepsilon > 0$ some term exceeds $L - \varepsilon$, and all later terms remain between $L - \varepsilon$ and L.

Only after proving convergence do we pass to the limit in the recurrence. Continuity of the square root gives the positive solution of $x = \sqrt{2 + x}$, namely $x^2 - x - 2 = 0 \Rightarrow x = 2$.

Thus,

$$\sup = 2, \qquad \inf = \sqrt{2}.$$

What a bound does, and does not, tell us

A supremum concerns all terms, including the first few; a limit ignores any finite initial segment. For instance, changing a_1 to 100 changes the supremum but not a pre-existing limit. Boundedness alone does not imply convergence, as $(-1)^n$ shows. Monotonicity is the extra hypothesis in the theorem used for the radical recurrence.

Review problems

1. Explain why the IVT cannot establish uniqueness of a root. Identify where completeness enters its proof.
2. Find both asymptotes of $(x^2 - 3)/(2x - 4)$, and explain why a function can cross a horizontal or oblique asymptote.

3. Compare $(-1)^n n$, $n^{(-1)^n}$ and $(-1)^n/n$: determine boundedness, convergence, infimum and supremum separately.
4. Prove the radical recurrence stays below 2 before attempting to solve its limiting equation.

Lecture sources

Xing-Bin Pan, MAT1011 lecture slides: Lecture 5 (continuity of compositions and intermediate values), Lecture 6 (infinite limits and asymptotes; the nested-radical exercise is on p.32), Lecture 7 (sequences, pp.3–33; completeness and Example 1.1.47, pp.34–41). Tutorial 1 supplies the proof-language section. Differentiation begins in Week 3.